

NFL Scoring Trends Case Study Rubric

DS 4002 – Spring 2026

Why am I doing this?

This case study allows you to use data science techniques to analyze how offensive strategy and game pace impact scoring in the NFL. You will work with real-world data to explore trends, build models, and interpret results in a practical context.

What am I going to do?

You will use a dataset derived from nflfastR play-by-play data from the 2010–2024 NFL seasons. The data has been cleaned and aggregated to the weekly level and includes variables such as average points per game, pass rate, and plays per game.

You will conduct exploratory data analysis to examine trends, then build a regression model to evaluate how pass rate and plays per game relate to scoring. Finally, you will interpret your results and identify the strongest predictors of scoring.

Your final deliverables include:

- A GitHub repository containing all materials
 - A Jupyter Notebook with EDA and modeling
 - All of the Data
 - Visualizations showing trends and relationships
 - A written summary of findings

How will I know I have succeeded?

You will meet expectations when you successfully complete the criteria in the rubric below:

Spec Category	Spec Details
Formatting	One GitHub repository ● Create a repository ○ Repository must include: ■ README.md ■ Scripts ■ Data ■ Visualizations/Outputs ■ Written summary
README.md	● Brief summary of the case study ● Explanation of the dataset and materials

	• Description of the purpose of the analysis • Clear explanation of the problem being studied (NFL scoring trends)
Scripts	Well documented Jupyter Notebook (and/or R script) that includes: • Data cleaning and preparation • EDA and visualizations ◦ Regression modeling using pass rate, plays per game, and time ◦ Model performance (R^2) ◦ Clear comments throughout explaining each step
Data	• Dataset used for analysis must be included in the repository and be clearly labeled and easy to interpret • Include cleaned dataset (e.g., weekly aggregated data) • Any additional datasets used should also be included
Visualizations/Outputs	Include all visualizations showing trends between scoring, pass rate, and plays per game
Written summary	Explain the overall findings and conclusions of the analysis. • 1-2 pages • Pdf format • Uploaded as a file to the repository Order: • Title • Name, course, and date • Overview of research question and analysis approach • Key results from EDA • Discussion of overall significance of results in the context of NFL strategy
REFERENCES.md	• Markdown file titled REFERENCES.md • Include all sources used in IEEE citation format w/ brief annotations explaining how each source contributed to the project